

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: Fe-59			FORMS: Soluble		
PHYSICAL CHARACTERISTICS HALF-LIFE: 44.5 days DECAY EMISSIONS					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
1100	57	149	53		
1292	43	81	45		
192	3	36	1		
143	1				
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY C – level (low hazard): ≤ 2 mCi B – level (moderate hazard): > 2 mCi, ≤ 100 mCi A – level (high hazard): > 100 mCi					
EXTERNAL RADIATION HAZARDS Gamma dose rate, point source at 1 ft, 1 mCi: 5.9 mrem/h Beta dose rate to skin, point source at 1 ft, 1 mCi: 130 mrem/h Contamination skin dose, uniform deposit of 1 μ Ci per cm ² : 3600 mrem/h			INTERNAL RADIATION HAZARDS Annual Limit on Intake: 800 μCi (Ingestion) 300 μCi (Inhalation) The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE).		
SHIELDING Gammas/X-rays: 4.4 cm of lead will reduce the gamma dose rate by 90%. Betas/electrons: 1.2 mm of plastic will absorb all emissions. Bremsstrahlung may be created and require additional shielding.			DOSIMETRY AND BIOASSAY REQS Whole-body and finger-ring dosimeters are required for handling 5 mCi or more, or 1 mCi amounts weekly . Urine assays may be required after large spills or contaminations.		
SPECIAL PROBLEMS AND PRECAUTIONS: 1. Recommended survey probe: NaI or PGM 2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often. 3. Survey work areas before, during, and after work. Work areas may require shielding to keep dose ALARA. Instrument and smear surveys are required. 4. Segregate waste to those with half-lives between 15 and 120 days. Survey the waste disposal area to ensure exposure rates are less than 2 mR/hr at 1 foot. 5. Limit soluble waste to the sewer to less than 10 μCi/day per lab.					

References:

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- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
- Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
- 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>